

ActInf GuestStream 050.1 ~ Valeria Becattini & Anna Ciaunica, "Selfless M

GuestStream | Valeria Becattini & Anna Ciaunica | 1:27:28

hello and welcome it's July 27 2023 and
we're here in actin guest stream number
50.1 with Valeria becatini and Anna C
Unica this is going to be a very
interesting and cool presentation and
discussion on selfless Minds unlimited
bodies homeostatic bodily
self-regulation and meditative
experiences there will be a presentation
followed by a discussion so thank you
both for joining and to you Valeria for
the presentation
thank you Diana very much I'm very happy
to be here with Anna presenting a path
of our

[Music]

um
wider project is a paper about what
happened in Southwest state from a homo
suffering body self-regulation
perspective
so it's as I said it's part of a wider
project that consists in both
theoretical and experimental
um

components and

So today we're gonna look at
meditation

so for people that are not really
familiar with that organization is
attracted engaged attention to
self-regulate physiological and
cognitive activity

and what we're interested here is that
people that meditate with a certain

experience so for example meditating in the past few years with regularly like some hours per week they can consistently report losing oneself uh one sense of the self or having selfless experiences but only not only that they also report to have uh bodily uh alterations so alterations in the procession of the body that has been called expansion Oceanic boundlessness dissolution of body boundaries and since you are particularly interested in this kind of phenomenal experience we have proposed to frame them as unlimited body just uh for clarity so our question here is what exactly and the question there was we'll follow the whole uh presentation in the background is what exactly exactly is lost when I experience many faces report losing their self and the body so let's first look uh very briefly uh what what um the self is or the sounds of selfies so this content is so complex and so wide that as we debated since the beginning of philosophy um and of course it's so it might be so intrusive that it's also very hard to Define but one very old uh old and used definition is um the unitary or unique individual I the first person perspective that laughing was called the subjectum so something that lay under or uh stream of experience thoughts and whatever else

happens to us
that's the subject
um however more recently in cognitive
signs we are appealing to a more common
definition of sense of self a bit more
exhaustive that is um defined as a
multi-dimensional construct and here uh
we will come back on this on this
concept later and we'll unpack it a bit
more
so the aim of this presentation will be
to first of all lay down the fact that
in self-experiences the biological
organism so to speak the the human body
remains successful that self-regulated
and this uh might be very intuitive but
it needs them to see that
during selfless uh experiences what we
propose is that they may be uh tacitly
a compiler yeah implicitly underlined by
enhanced connectedness with this
successful self-regulation that we
proposed to Lincoln with a perceptually
attenuated body or what has been already
called transparent body
uh and this uh allows it to
propose it's of course in a working
process and we're very open to uh
feedback at any stage of this
presentation that the term selfless
States may be misleading because what is
lost is an ordinary way to mentally
models the self relation to the body and
the world
so I wrote my is composed as uh as
follows
first I will um
passing through uh debates and

definitions of losing oneself and body
so that means they'll go through
alterations in these um three-dimension
uh during that deep massive state
and then I will introduce King concept
of predictive process interactive info
framework
that are
useful in our explanation
so that then I can uh
um unpack the role of the biological
self-organization and homostatic
self-regulation in constituting our
sense of self
and then Pastor or uh claim that the so
much process is like some other sensory
attenuation and transparent self
modeling our key mechanism underlying
the phenomenology of the Methodist
States
our conclusion and present uh ongoing
related projects
so let's back come back to the sensor
itself
so uh in the
um
in when one has to uh investigate the
sense of self of course it as I said
it's a wide concept there so many
definitions and
um characterization of the sensor cell
has come up uh in philosophy and
cognitive science so we have uh terms
like minimal self self experience
self-hood sense of ownership and all of
this and
for author I think we have just to stick
to a common definition

that comes for example from from a similar paper from Gallagher to thousand that defines it as a multi-dimensional construct composed by minimal or embodied components and narrative or reflective components so to unpack a little bit these two concepts the minimal embodied experience of the sense of self is the basic um implicit functioning of the um by the YouTube of the ongoing moment so the stream of experience that happens uh to me right now uh can be defined as my minimal sense of self so I don't really need to have an explicit knowledge about myself to have this kind of Dimension uh and these uh can be further unpacked through the accounts of our pre-reflective self-consciousness that comes uh from a phonological tradition tracing back from uh about to uh uh argue that uh parts of our self-conscious does not does not really need any reflection to to happen there or in place an explicit thought it happens um in a way tambrarian uh is conservatively before uh reflection and then on top of this we build our narrative or our Reflections about ourselves and this is more linked to uh self-identification processes or self-monitoring self-recognition and introspection introspection it's it's um the ability of introspect so perceive yourself or reflect about yourself in an explicit

way

so you do need to have a biography or an explicit memory about yourself in order to tackle this this aspect of this answer so

so indeed massive States uh the there is alterations in sense of cell and it's up to debate what which um Dimension are affected by that and it's far from Clear what um actually

happens to these kind of Imagine a more and many more other dimensional sense itself uh and we are here uh to unpack a little bit this

we propose these quotations from Fashion 2008 uh where

um it suggested that one in doing deep messy stage one experiences oneself and this very moment of Consciousness this is a quotation that may reveal some uh position about this

um the debate what happened to the sense of self so in this case it's disargued that meditation reveal a very moment of conscience so there is something uh that is true consciences

or uh something that is the real Consciousness and this actually has breaks to the concept of pure Consciousness so something that it stays there and uh

um it's

um purified by everything by almost to the body and there is so many uh definition of this pure Consciousness we brought some example like uh ownerless Consciousness the conscious that doesn't

have an owner that doesn't have a self
so it doesn't have a proper sense of
self that walked or that owns this
constant uh nothing it has worked on the
minimal phenomenal experience defining
it a special a temporal dimensional
consciousness

and this is uh very supportive because
it's consistent with the Buddhist
philosophy philosophical standard of no
self so for the uh most tradition of
Buddhism the
um there is no self the self in is an
illusion and it comes uh to our
psychology when we enter in contact with
the world and with the sensations of the
world and certain uh patterns of
behavior the analysis the tension we
have a self

so as it's that many patient it
allows us to access this pure
Consciousness for for supporting to this
thesis and reveals one to choose an
essential nature or subjective and we
bring this into debate

uh

there are also uh people that argue
against this and it's called Uh it has
been called by Imagine their
special issues the ubiquity thesis
so the ability thesis claim that
essential cells is ubiquitous to all
states of Consciousness and sometimes it
appeals to the immunity to error through
misidentification

argument of self-ass position reports
which is simpler than this long name
um and it's basically

the fact that I can say that I had a wonderful experience or I can and this wonderful experience may be true or false this is uh this on this weekend uh I can do mistakes I can make mistakes but I cannot not say that I didn't uh had this wonderful experience because the moment in which I report uh I have a perspective on this so I cannot really be mistaken on the fat end reporting this maybe I was dreaming maybe it wasn't a wonderful experience but it's me there remember party so the moment I'm reporting this it means that I have to be there and this is uh for uh applies also to uh when I ascribe myself a selfless State I had a self-estate this this is very controversial other problems on top of that are cross-cultural problems also addressed in the leadership so this experience selfless state may be influenced by uh certain vows or certain beliefs that meditators are endorsed uh especially in during especially expert meditators and the other uh thing is uh these uh all um um cross-cultural analysis and debate about yourself maybe also uh actually due to our verbal misunderstanding so that the western western employee may have problems about the self that actually the Asian probably philosophy is not really addressing so there might be some

incommensurability of of problems there
but as also we said in the in the paper
uh it's not there we want to address the
Buddhist philosophical problems but
rather we want to bring inside from from
these two to our to our problems

so the

um given this debate uh we want to
propose a shift in the focus so we don't
want to start by looking at

Consciousness and trying to solve things
about the self but we want to start by
the by focusing on the body so we
proposed a shift in focus in the bodily
experience examining this uh the
imaginative state

so

let's start by the subjective experience
of the body or the study of this ability
to experience of the body which is
phenomenology

uh in the tradition our body has a
unique stages

so the anatomical structure of the body
uh has been called by user cup so the
living body the body that I can observe
and others can be observous is the
object among the other object they would
say

but it's not only that because

uh it's an object that I can live from
inside so it has been called the live
the live body or the body as I leave it
from inside as I said

so what happens uh to these kind of
Dimension again we can unpack this and
see how the meditative States affect
them

so from the point of view of the life
believed body uh we already know and I
will unpack a little bit more so we have
already said that uh we have called
unlimited body experiences uh the three
parts that spoke about this
um fading of the body boundaries so the
body bungee has been defined as a
cognitive emotional perception of uh
one's body surface that just finishing
we can also debate about this but well
it's important for us that our sense of
body boundaries is flexible and dynamic
it's not fixed because we are in the
realm of the body as Fells from inside
there is an interesting study that
exemplifies this concept in relation to
meditation in its uh by John brand in
2016 though it has worked uh late
further on on this um as well but I find
it interesting to look at this image
because

this is part of a self-report management
that they use on a sample of people that
has been trained in meditation in
particular with the body scanning so
that people had to systematically scan
sensation rise in the body
and the result is that in the less than
four months of practice so no actually
not the expert they already started to
report an increased tendency to go
towards the
almost imperceptible
perception of the body so they conclude
that um meditation can be actually
studied in this sense where it affects
our sense of body boundaries and uh they

say they saw significant uh results also
in relation to increased mood happiness
and
decreased anxiety and regulation but
this is uh yes another story that we'll
come back later
so I find interesting also this
quotation by I think uh here in this
paper is talking about why in good
dissolution may be better understood as
ego expansion and then
um and then either the solution becomes
an a later uh
interpretation
um and so I will read it it's it's not
the feeling of self that dissolves with
the boundary between a feeling of self
and the feeling of anything else in
consciousness
so here again I think he's talking about
consciousness but
from our perspective this conversation
is pretty uh informing because it goes
in our interaction as well because
he says it's not that you're losing
really yourself is this this experience
is uh dependent on how you actually
perceive your boundary with uh you and
what you don't consider you so
self or fading of the self other
distinction ability
so
here it comes uh processing acting from
framework because we want to
um
basically breach these phenomenal
subjects you experience to having this
phenomena and underlying mechanics

explanation

uh indeed the processing activities I have recently used for this purpose also meditation research and uh Psychopathology research and so on so in this presentation I will use them in an interchangeable way but following certain uh Trends but it's not it doesn't have to be it's also under debate

um and I will also give if I uh purely I don't know how to say

um theoretical

perspective on that I will not go too much technical

um so briefly uh this framework considers uh

brains as predictive machine or as a freezing likes to say as a considered brain is a fantastic organ uh which is because not it's not only uh able to do amazing things but also because it really generates fantasies about the world and this is because the in order to act interact with your with Excel and the world around the brain constantly uh anticipate

um

the few anticipate the state current state of affairs

in order to make sense of the sensorium or the stream of experiences that happens

um so on uh so it possessed or embody or uh utilizes it depends on the commitment uh a generative model which is a model that keep tracks or regularity of the world and forms prediction out of it in

a hierarchical way so highly
reproduction uh goes down to match uh
sensory data
um and sensory data goes up to uh
um map to affect the predictions to
update the model
however every prediction comes with a
degree of error which sometimes is
called prediction errors so
this predictionaries the discrepancy
between prediction and
um sensory input so that a new model is
generated and uh they say the
um first imperative or the only
imperative of the system is actually
minimizing the prediction errors so that
the model can most efficiently uh cope
when interacting uh and act in the world
however it's not um
the only imperative because the as I
will introduce in this Slide the
the system uh has also to
estimate the confidence or likelihood or
Precision of prediction in order to
evaluate the prediction errors uh that
are coming from
the discrepancy
so that the generative model has to
compute enough a fine way uh how how
much Precision to assign the to the
prediction or prediction errors
and I choose this image because it
renders well
um
how the distribution are
um set depending on the prestigious
sedation so if you have a precise
medication error

then the posterior distribution will be closer to the prediction error well the imprecise prediction have lethal effect on the prior so that the procedure the the pursuit distribution will be closer to the prior and this is very important because if the um that Precision estimate are not weighted correctly then the system generates false inference that will not able to the system to interact efficiently and update model efficiently so so far we have uh mostly talked about uh prediction error minimization uh by updating uh the model however the system can also uh active samples or active modifying the prediction error of the world um in order to fit uh the prediction errors to the world and this comes into play active inference um that builds upon the free energy principle where uh that um is based on the idea that living organism maintaining themselves with an optimal State set of states that comes from the second law dynamics of shortening Schrodinger um so given that it comes into play uh how attention is uh analyzer examine it under this framework so given that uh the organism has to maintain uh Within These optimal assessor states by maintaining an optimal balance between uh computation

uh and precision optimization SO waiting
between uh prediction and
um prediction errors attention comes
into place as a modulation of these
Precision of sensory data working as a
Searchlight so in other words it
optimize the signal to noise ratio it's
the one of the way that we have to uh
put in place in place these local
optimization of precision estimate
and this is very important uh why we
bring into attention because meditation
has a systematic energy
reservative use of attention resources
um

and we will come back into that
further on in this presentation
uh and given that we are interested in
okay we have seen the live body what
happens to the funeral experience now
we're interested in what happens to the
corporate what happens to the living
body that can be observed by the the
organism that during medicine States
in these bring we brings into play in
other two uh important concept uh in
active immigrants which is a way to uh
self-regulate

um
the underlying all and what experience
Soul subjective experience
so two ways uh
or one way that then is also active so
the first way is homos little homeostasis
which is the maintenance of required
optimal Say by survival and reproduction
of self-organizing biological system so
for example I'm cold and I use my

internal resources to keep my
state into this into optimal state so I
start shivering however I can also put
in place uh active modification in the
world so that
um I'm not cold anymore
so for example I take a blank this is
called elasticity so it's active
engagement and modification about the
environment in order to maintain myself
in optimal States for survival
so we can say
uh of course human bodies are
self-organizing biological system and
aiming at keeping these internal States
within a certain uh viable range for
survival
now given that we will come back again
to phenomenology
off touch I will go I'm going to to do
this for the second time
uh to then wrap up everything again
uh so we have talked about film all
video of the body
however uh we haven't uh underlined
important is that the touch and proximal
sense has for our lip body
so
our as Technica and colleagues has
already
um
proposed our proximal sense are Central
for our minimal sense of self
and on this on on top of that we bring
important to play the fact that
different kinds of touch can convey
different experience of this and so
stuff or sense of boundaries so for

example imagine yourself that you are
um in a c
of partial or California and
you touching the whole environment with
your body
you may feel that you have a sense of
immersiveness or that your body
boundaries are
slowly a bit becoming less clear and
this experience it's actually a bit
similar to the experiences of sensory
deprivation States where people are put
in a salty water and they let them flow
for some moments
sensory deprivation states are not
really really all deprived because you
have other activities in the body and
movements that produce signals that you
are actually aware of
however in both these experiences the
body boundaries seems to fade a little
bit
however if you're in the cnu's meeting
and then you touch the rock with your
hand uh you clearly will feel yourself
as separated as uh something different
from what you're actually touching like
sense of separation
things phenomenology can be both
[Music]
interactive inference through the
mechanism of some of the sensory
attenuation which comes from the sensory
attenuation
so this process happens when uh the
system depicts your its own action
uh so for example I take a glass
uh and in this moment my Rena has

already created a different copies of
this action and the sensory consequences
of this so whenever I grab the
um these this class the sensory concept
that happens to my uh but they are
slightly
um attenuated
and this is why we cannot tickle
ourselves
because the prediction errors comes at
not surprising
this has a implication for our embodied
experience in the world because we are
feel that we are immediate immediate
content with ourselves in the world and
this is because
our country experience are mediated by
transcript or non-perceived cognitive
processes so that the system leads can
leave behind or this attend self-related
sensory processing
so and these allowed to allocate your
resources to perception and action
involved in the processing of the object
of Interest
so for example when uh limousine as this
example of the Cherry when I have to uh
take a object that Italian for me I'll
be much more interested in the uh
picking the object and I will not really
focus on my arm boring
in picking that
so that the system needs to compute and
maintain a fine balance between external
and internal comp components of some
other sensory attenuation
and this is very important for the sense
of self as well

so it uh we need to understand and
result that we need but in order for us
to have a sense of self we normally have
a sense of what we are when we don't and
what we are not but for the brain it has
to uh or as as uh agents end up with the
brain we compute this distinction so
where the self ends and where do work
begins or do I move my arm or my arm is
moved by something or someone
play the role in this uh distinctions
and involved in building multi-sensory
bodily self-representations
so that the sense of the self and body
is uh subjectively and phenomenally
transferred that's that that means that
when we act in the world and we interact
we normally these attend uh the
way we are in the world because we need
to be in contact Direct contact with the
world if we always attend it to our body
and our our
um ourselves in to in the sense that we
will be hyper reflexive then we will be
actually uh will be impaired
in what we normally do and this is
actually a condition that we will talk
about later
so that
in our students in the sense of selfies
and ongoing inferential process center
in the body so what we propose is
uh what is lost during selfish selfless
experience is an ordinary way to mental
mentally model self in relation to the
body in the world so that if we think
about the self as as I propose here then
I uh the iceberg where the narrative

cells is what we see but the renewal
self is what we actually are composed of
and it's actually larger but we are not
aware of similarly when we have software
experience we may have also these uh
inferential process where we think we'll
have lost ourselves
um however
uh we have gathered some evidence in the
fact that meditation facilitate all
genomic regulation involved in
homostatic processes
that means for example that meditation
increase the synchronization of cardiac
and respiratory system
uh this is connected with reduced levels
of blue blood glucose or blood pressure
and heart rate variability so there is a
um
successful medication helps to
successfully self-regulated the body uh
especially in comparison to stress that
actually does the opposite
um
and on top of that uh meditation help us
to attune us to our physiological
interceptive awareness uh and this
um it does it so they say by enhancing
the processing of extra effective
interceptive stimuli which it doesn't
have to be understood as a maladaptive
uh increased body awareness but rather a
adaptive body awareness which is very
distinguish distinction that we can come
back later if you
if people are interested or they ask
about
um

these evidence are also in line with
evidence communication that promotes
self-regulation of pain and pain and
pleasing and so
three-day attentional training the
meditation
in allow uh you're actually able to
self-regulate the pain and not not
really like the sensation of the pain
because it actually gets very clear but
the uh automatic behavioral patterns
that comes uh in association with the
with the pain
so there is also also self uh regulation
on emotions and so on
so the body doesn't know literally
disappears actually it stays as a very
centered uh as a first year at the
center as we propose it in the paper and
I proposition through this image at the
center of a cycling
so self-regular
self-regulatory processes allow the
subjective experience to lose oneself
so the percent of the body
remains present in the background and
these for example can it's very here
from these reports from a
person undergoing this kind of
experience that they uh say that there
remains a sense that there is a body
without any experience of a sense of
boundary
so
this is
not very last slide but we come to our
wrap up of our claim
uh attenuating the self unlimiting the

body which is also the title for our paper so starting from the fact that all medicine practices across the three Traditions require systematic and parallel fully crafted body preparation to enter these deep method self-state self-estate so it doesn't happen that all of a sudden you just have a selfless state which actually may happen more similarly with a psychedelic bubble meditation you need you need to pass through gradually many different uh state or many of different um level of concentration that happens when you have around a predictable environment and you do so by building prior expectations so you have to keep your body in a certain position of posture as you say this in this image the person is seated with a very straight but not very but a straight back relaxed this is very important realization when it lags and so on and then you need at least uh in in the most part of your practice you need safe predictable environments with temperature that are enables you to maintain your homostatic analysis and processes so all these conditions are necessary for your successful homeostatic self-regulation in this sense when this part first part

of the preparation to Andrew then after
the state of concentration the meditator
can let go and leave behind self-related
processes and process the body
transported or it can attenuate them
just attenuate its body the body cannot
can be attenuated
so our claim is that a transfer
essential cells and body May underline
the phenomenal phenomenology of losing
one cells one self and blurring body
boundaries as reported during meditative
States

um

so our conclusion is that we have
proposed a shifting focus on homeostatic
body self-regulation in examining the
majority States

the experience of having ability a
living self-organism biological system
is never lost in this process
rather it stays transported at its very
center

self-attenuated yeah very much present
in a handful life

so that the self model is attenuated to
a big week to become transparent
enhanced processed in the background
now giving this conclusion now I'm gonna
present related project the first one is
still uh on the theoretical side so we
are in the real of active inference
model in meditation research there is a
radio literature about that

um that

um base that have studied different
kinds of meditation techniques like
focused attention open monitoring

non-dual awareness

2021

um they have for example uh showed that these uh attention techniques profoundly and alter typical mental experiences and

for example Focus meditation is the meditation

um that engage attention on one specific object experience for example the breath uh and in active events there we mean that Precision is uh estimated one century input which means that it reduced the impact of the other thoughts and parallel processes that going on or often open monitoring meditation is the uh is the engaging of attention to one's experience as a whole so the whole stream of experience with us selecting anything in particular so that in an active intertherens it has been described as assigning equal Precision to all these experiencing promoting non-reactive sensing and it is connected also why meditators

are good in self-regulated regulate pain so in we will also build our model our us on self modeling based uh on the previous literature which is in preparation

and then going experimental uh we want to bring into the table uh applications of what we we're researching and potential benefits may be for a case or people or yeah individuals that suffer from disrupted body awareness uh or disruptive self-awareness and there is a

relatively recent condition that has
been
studied that is across the
personalization
so the personalization
um is their destruction of both body and
self-awareness and it's a very
heterogeneous condition there is a lot
of different dimensions uh that occurs
but generally uh it's connected with
feelings of Happiness disconnection
feeling the body isn't the as an object
rather than a transparent medium of
experience uh which is uh switching
release called disembodiment
it is linked with difficulties in
concentration and attention often
comorbid with depression rumination and
anxiety
and recently and colleagues that also
propose a uh characterized by failures
in attenuating so much a sensory signals
so
based on what we have uh previously said
and based on how these
um condition affects uh
a large number of population we want to
test certain hypotheses
so the first one would be the highest
somatocentric attenuation correlate with
more feeling of aliveness and wide
openness in experienced meditators
and rather lowest from other Century
attenuation could relate with feelings
of unrealness Badness in people
experience uh decorationalization
and on the other hand severe homeostatic
disregulation or bodily States during

deep Methodist States may lead to negative emotional outcomes and other self-experience this comes from to the fact that

the personalization and meditator and meditation has some has uh can both create stressing negative outcomes and it's interesting to see how these happen and what are the commonalities and differences in that

and also we have now in the realm of spatial temporal perception which is uh connected also with mood disorders or like depression and anxiety uh and also with the personalization

uh however in the precise station there is no specific items in the questionnaire that is usually uh used for measuring so we're which is a Camry Cambridge the personalization scale so we want to test this as well uh and we want to test it uh

in people reporting selfless and disembodied experience so both pathological non-pathological conditions so path of this hypothesis we are already going through

this is our team coming from Humboldt University where I come from the month but also a university Hall and College London with anus and Joseph

first study is as I said about time perspective so we want to analyze uh um how people with high rates of the crystallization can relate with dimension of the time perspective so there are more future oriented or more negative oriented our hypothesis that

they're more negative past
oriented so they're stuck in
a very
long present moment that but also the
the their the tendency of ruminating so
always going to pass negative memories
um

this is a title lost in dark times the
effect of the personalization
experiencing on time perspective
and these we are already in the analysis
of the data but

um
it's on ongoing still
study so we we have preliminary data on
this

uh while studies two we're in the
recruitment phases but we are almost
there and encompassed the
uh hydrates or people individual with
high traits of the personalization and
meditators

I want to see through a set of questions
and a behavioral task how they may
differ in the sense of self body and
motion of course they're very wide and
conflict Concepts so there is always
room for for
feedback and
replication and so on uh we want to just
we wanted to we have selected this tasks
the embody task that I've adopted from
newmanma

uh where individuals have participants
has two colors parts of the body
um

in relation to activation that they feel
by reading certain emotional story

so these stories at least are supposed
and validated to at least see certain
emotions in the body and then people
colors parts of the body and then we
will see how people with high traits and
visualization limit heaters will will
different this and we will also compare
with other questions that we have put
this and

uh potential title is emotional body
maps of solid Detachment personalization
and meditation

and with this I thank you all and that's
it now we are open for discussion thank
you

awesome thank you very much Valeria
your music down

thank you Valeria

great of course well

very fascinating presentation perhaps uh
on up first and then our other
colleagues who have joined would be
awesome to hear your

after

be here on this work and and any
reflection or questions you have on it
is it my turn sure did I missed yeah
yeah so um obviously I don't have many
questions because it's like we
brought this together so

um

but um I I think what I'd like just to
add that really a question is

um

I really like to see

um

um so for instance like in um an ideal
world would be nice to you know uh get

all the

um

the tradition of the meditative
experiences and how they relate to the
body and make a very precise prediction
on what exactly they will kind of like
predict in terms of uh certain

um

uh sensory experiences and how this can
be interpreted through different like
theoretical lenses I think that could be
really really interesting because I
think one of the problems
uh the challenges that we have right now
is that we have so many Concepts so many
approaches and kind of like scattered
all over the map and I think would be
really nice to

um you know a narrow down because as you
said Daniel

um so everybody knows what meditation is
everybody knows at least one person
meditates in in life and everything but
uh it's really unclear what exactly we
are um we are after and why in certain
cases this is kind of like

um beneficial and one in another cases
basically it has a negative consequences
and this is why precisely we are doing
this our project on the personalization
and meditation and I can share with you
informally scoop but data is here so it
kind of like I can share with you so
what we did in this project basically we
took people with typical controls that
don't have the personalization and never
meditate it right so it's kind of like
yeah never did that and then we took

people that meditated but
um and then we looked at whether they
have kind of like high or low traits of
depressurization and then we took people
with deeper sizes that never meditate
right

um so

so I have this like

um inclusion in exclusion criteria and

um yeah it's it's really interesting to

find what our finance is like it's

really difficult to find people that

have experienced meditators and low

depersonalization skills low

depersonalization traits right so most

of them they actually have ID

personalization but then it's

interesting to see how that contrasts

with like the people that have the

personalization and no meditation

experience right

um so that's that's something I would

like to

um Carry On and

um yeah so to reach the community and

um yeah

and that's something we will definitely

carry on with Valerian the next steps

and also with some people that I have

here in the chat uh Julia and Simon so

Simone will going to work with us on

also smarter sense retaliation yeah

through different senses and Julia we

have some um without scooping we have

also some very very interesting results

coming up

um we just finished data collection on

intentional binding and

depersonalization so related to the sense of agency yeah so yeah so that's all from here just like for now we're kind of like in between because well I represented very well like the theory part and the studies are literally just getting hot from the oven now and we hope that in September it's like we're going to get the mode there yeah but the data is in the box so that's very exciting and yeah we have the results yeah

thank you Danielle again for uh this platform it's really nice and as I said I really hope that after this discussion we'll have more

um um you know

opportunities to change exchange and um yeah discuss we also plan to organize something around a duration of self-consciousness in across various conditions and also the personalization in Elizabeth next year so maybe you can cross the pond and check for yourself that we have a nice airport in Lisbon yeah

cool well good to uh get the appetizer before the entree so that's very awesome about just how much Theory and um even unconventional scientific theory like World Knowledge traditions come into play and then what it looks like to take that experimental step to make it scientific is is definitely what I

saw on the talk

um Julia or Simone do you want to give any reflection or any thoughts or

questions

yes I actually have a can you hear me

yep yeah I actually have uh three

questions I know it down

um I think the first one because it's

just for me personally I'm curious about

that is that those deep meditative

selfless States I think that can also be

reached while doing yoga right

basically the opposite of like sitting

still is like a sort of repetitive

movement actually so I'm wondering do

those movements have to be perfectly

predictable in order to be attenuated

and then reaching the same state or is

that like a different uh different

concept or how do you how do you put

that in the in your in your framework

um

mainly to

uh to talk to topics one is is it

possible to reach deep Methodist States

also uh not sitting in a steel position

and the answer is yes

uh

however there is not so much research on

that so I can talk about anatical

knowledge

but I did not read so much about uh

World communication effects

usually works meditation or yeah of

course you talk about yoga I know but

that's what I uh would say that walking

meditation is used uh as a complementary

practice through the same thing

so I would say

hard to reach deep Melody stage only by

World communication

um

in regarding to yoga also yoga means

Unity so what we think about yoga now we

think about movement but it's always

uh occupied by reduced state of

conscious concentration

through meditation

so you cannot have yoga without

meditation otherwise we'll be stretching

your glasses

but uh second topic

is that can it happen when you have when

you don't have prediction of movement

right no I meant like because in yoga I

mean at least I don't really do yoga

right but I can imagine that what when

people reach those States doing yoga

then they kind of do it almost like in a

in a trance type of way they're always

doing the same movement over and over

and over again so that they don't have

to attend to those movements but it's

just like an automatic process I was

wondering if it if that also can I can I

jump in so

um um actually this is precisely the

opposite what is what what Valeria was

saying is so but I was saying this is

precisely the opposite you want to reach

through yoga just like the opposite of

automatic Yeah so basically when you do

this movements you need to be United

with your self-training it so you really

need to

um yeah uh so yeah I'm doing yoga so

they they the past 10 years and

precisely this is like they're trying to

tell us it's like don't don't do it on

automatic because it's like when you have your thinking happening there and your body is doing some stuff there right you need to be basically be presents in every single type of moon so you have like a fully fully a presence if you do it automatically that's like that's more like gymnastic or something yeah so um but it's interesting uh so your point is really interesting because then would be really nice to have a control condition where we have people doing like movement meditation and yoga and then um contrast with the phenomenon of like

um uh some other sense retineration and uh whether they have what type of light embodiment and disembodiment adjectivity experiences they have

yeah

okay but yeah I don't know exactly what I was trying to get it but I think it was also more related to because you said that when meditating you have to sort of

let yourself

letting go exactly the ideas then everything has to be perfectly sort of predictable right that's that's the idea behind it so then I I'm I'm thinking so you always within yoga don't you have to do like the exact same movements in order for it to be predictable in order for for you to let go and not over attend to the environment does that make sense or

if I thought when when were you both uh on the Simo we're speaking I think on

one side you see more maybe you're referring to the implicit procedural map implicit procedural knowledge that can uh you can acquire through yoga and you and I are talking about doing things without being aware which is also automatic and left and you both are right I think in this because on one hand if you and you would say like if you do yoga you learn how to to do yoga right so you learn how to uh um move yourself in this in this way so that when you do a certain position you know then you have to arrange your body and for a ranger that you always have to keep monitoring yourself so there is a continuous going back to your body keeping track of your body and continue to monitor yourself but in this way because you have learned to do this these um constant constant going back and being aware on yourself and these comes also when you you do your movement you have learned that you certain posture are made in a certain way so that it becomes more or less predictable actually what you're gonna do right but it doesn't mean that you you don't have to be aware and actually there is a trick uh reload here because you're a very attentive with what's going on but at the same time the more your attentive the more so much a sensory attenuation may happen and actually it's what we are trying to argue in our active uh inference model the the more a predictable it gets the

more it gets attenuated but doesn't mean that you're not aware that's that's uh yeah that's good thank you for this input because it helps us elaborate this I think it's super interesting because it's like when you when when you look at like people that climb or surf or whatever any type of sports right in the beginning in order to learn the skill you have to put a lot of attention to the individual movements but then later on in order to actually be good at it you have to take the attention away and then in those moments is when the people report that they're actually feeling sort of selfless days in this flow States right which is when the the movements are internalized so to speak I think that's that's super interesting can I can I just shoot the other two questions I had they're like sort of different or should we go to Julia first maybe it's true yes I don't know I sorry um yeah sorry for the lighting and sorry because I'm in the lab so it's probably not the best position uh thank you very it was very interesting and I'm really uh looking forward to um yeah look at the uh yeah the results you you have in your study especially yeah I mean since uh as and I mentioned we had something interesting with the international Banking and the personalization and um yeah kind of happy about that and what I do so I'm really sorry because I listened kind of doing something else unfortunately in the lab I was busy um while you were

talking but I kind of like the idea of

um

yeah that's the self is attenuated or
not or increasing and kind of goes back
to just Simon the last comment from
Simon about expertise

um

I think in terms of motto uh interaction
there is some literal tribality
expert I was trying to Google that
um that I attended a workshop last year
and I think it was about once you're an
expert

you at the beginning you have this kind
of flexical you you you have a better
understanding Consciousness and then at
the end you kind of learn to let it go
to write the flow of the motion kind of
so it's not that you are and I think
it's probably maybe the same I don't
know whether yeah but attention in
meditation is that you have this maybe
more like not focused

uh it does not stay but more like open
um sorry I forgot the name of the
meditation type of mission you know but
open meditation and monitoring yeah yeah
exactly so you I mean I don't know it's
uh so it would be interesting and I
suggested to see what kind of meditation
what type of meditation maybe change the
sense of the self maybe at the beginning
it's more acute because you really
scrutinize your sensation and and then
after I don't know this is something
that you plan to do right or
should I go or do you want to go on
uh the

the first question was I know ah there is a body of evidence they actually say that in expert meditators uh attention to start decreasing but there is also because you allocate less and less attention but I would be careful with that because with with this because they say there is an uh decrease that uh decrease activity in anterior singlet cortex so they make inference about it um as as a meditators as well uh and having to know other people there are more experienced than me they would never say we lose our attention to that because our experience is so rich our Sanctuary signals and related thought process that you cannot afford to not be attentive however we can drink into the play different kinds of attention as you said there is focused attention or there's an attention that requires more explicit uh thought about that some more explicit kind of uh way to focus on things and there is more like a bear bear attention so when you focus on the whole experience and you let things go with a non-judgmental attitude and more open attitude and is actually this attitude we're very stressed in meditation techniques that's why they're a bit under lock and in research I feel because it's it's a lot about how we use attention but not the intent attention but not how we use intention so our attitude has the attitude that mediters has to be very open and very

non-judgmental and these reduce the
non-reactive incentive for for example
and reduce the ability to self-regulate
the your experience

your life yeah I have

partially answered

you know I mean I kind of like the idea
that you you linked that with attention
and um it's just I think maybe it will
be interesting to look at different type
of attention because

um

whether you know you have like so maybe
there is a difference between I don't
know different type of meditation May
develop different type of I don't know
if maybe the open

um meditation

he's more prone I don't know because you
mentioned the Instagram cortex into me
it's more associated with a certain type
of attention like more like maybe

um more prone for like a feedback

um or negativity or these kind of you

know error monitoring process so it's

not like your attention you lose your

attention it's just it's more like in

the background and then when there is a

mistake or something it kind of pops up

more quickly I don't know and you may

have more like uh frontal or a sustain

attention I don't know I mean I'm not

especially of attention at all so please

take that little bit of self

um yeah I was wondering if this is

something I don't know obviously I mean

because it's only you're not trying to

do any EG or or narrow physiological

measurement right

yeah but in the in the pipeline yeah

okay but we really want uh with um with

Valeria is

um

and other collaborators is to basically

push against this idea that uh you can

have selfless experiences I know it's

controversial and many people talk about

oh selfless experience below

uh so basically we want to say radically

speaking okay though so that works if

you buy the bullet in your mind what you

do a list and somehow consider that

actually the body kind of like

disappears in the process but if you're

not mind body dualist then you need to

make basically put the body into the

equation and realize that actually what

allows the mind to disappear is the fact

into the pureness of the experiences the

fact that the impure body is kind of

like put in a

stable condition in such a way that

basically can afford to live it in the

background and yeah to have like this

type of like um experience arising but

without that in place uh you it can't

cannot really afford to you know go

through the selfless experiences and

this is why we want to be provocative in

the titles that well actually maybe the

experience of like selfless

um you know selfless experiences

actually comes

necessarily as to sign of the same coin

of this experience of like having a

unlimited body that actually expands to

through the world but the body is there
I mean the the thing the The Entity that
has the experience of no experience is
that a body that kind of like is
um self-regulated in a certain way like
the homeostasis is put at like a set
point and when you're using this
metaphor of like think of um
a cyclone Right Moving Cyclone so you
have the steel eye of the cyclone in the
middle doesn't mean that actually is not
there it's just like it's around
everything and because we are
self-organized in biological systems uh
we kind of like this animal this thing
needs to keep track of this internal
bodily states and their I mean just a
successful system does this all the time
all the time literally from second one
when we basically get from one cell to
you know the substanceism some sort of
like fascinating self-organization that
is happening on the cellular level
already and it like needs to be yeah and
death is basically this is what is
happening it's like a single
disintegration of like this
um you know self-organization and um but
that's clearly that's not what is
happening on meditation right because
you do hopefully you don't die when you
meditate in that sense there is a
disintegration of a certain way to
experience yourself in relation to the
body in the world but clearly there's
something that stays there needs to stay
there for you to have those experiences
so that's a bit of controversial radical

claim that we like to defend and uh yeah we already had this discussion several people for instance like whatever and I we've been to uh Enlighten and this workshop on um of knowledge and neural computation we met there with people like Thomas betsiger and everything and they always have a different opinion so kind of like have a lively discussion about why we think it's like what does it mean to actually Lose Yourself and what is the self and what is the pure experience and why we basically need whether we're like it or not the connection with the impure body to make sure that it can have at a certain point of pure experience but then but then again this is something that just speculation at the theory level and this is why I have this brilliant team here that we we can do some uh studies and actually measure stuff put the finger on it put the number yeah and see what we get these are these are awesome points and there's so many ways it could go I'll read a question from the live chat so G pagnani wrote the notion of increased somata sensory transparency in experienced meditators associated with increased liveliness of the experience is intriguing can you say something about which levels in the hierarchical structure of the internal model are more likely to be affected by the meditative attentional tasks in terms of precision deployment do you

think the attenuation of somatosensory
prediction error concerns more
intermediate levels in a way increased
meta awareness rather than sheer sensory
attention

I just wanted to ask rather than
ensuring
but rather than sheer sensory attention
yeah you can go on now please yeah so uh
um

yeah so this is something I I'm I'm
trying to develop this idea right now
with
um

um a PhD student of mine
um uh Libby uh Liberty service basically
uh and this is why it's connected with
the question that you ask you'll see is
that um

you know your platform is active
inference right and so people are kind
of like you know um

lately kind of like realize wait a
minute actually we're active and we need
to explain action in the world and
everything is kind of like comes late
right but uh if you look uh going back
to the animal basically

um the action and the movement is there
from the very beginning so this means
that this is why it's so important to
basically relate the um um you know
self-organization self-awareness
self-perception necessarily related with
movement not so movement so on action is
not something that you add on the top of
this right like uh and some people like
intention so you can have an intention

but basically the first movements are kind of like automatic and need to be there right and basically our body moves all the time because we breathe all the time and our heart beats all the time so this means that for me this type of like attenuation it's happening at the sensory level because that's the basic one so can you have sensory stuff without metacognitive stuff yes can you have the cognitive stuff without sensory stuff no so it's like what what is basic that's the first things needs to be um um uh alternated and then then it has to be a balance between meta cognitive and sensory attenuation because uh this is precisely what I think it is happening in the personalization here because if you if you focus too much right so if you lose the the balance right between then you you basically um

yeah it's it's kind of like this type of you cannot afford to sell forget yourself yeah and then it's kind of like the self-standing way and that gets to get you even more disconnected so to speak yeah uh so this is why I think it's important to um uh contrast different type of like a meditative experiences and meditative practices to see which one of them actually um has more focus on what you do with the metacognitive metal Oneness awareness of awareness and which one are actually involving more like sensory movement for instance like let's say

yeah so that's that's something that
needs to be um empirically tested but I
would speculate and then obviously we
can put a test either by doing a
simulation or you know actually
collecting data to see how that
um uh plays into this uh distinction
I think Simon wants to say something but
I'm not the moderator So Daniel you
decide

yeah Valeria first and then um
okay uh

yeah thank you Anna that was very uh I
don't have so much more to say because I
agree with her but uh

I thought when Jani and you were reading
the question I for me the answer was
more or less clear in the sense that
um as I say the somatician continuation
happens are the very low level
predictions so

in a very uh pre-reflective
self-consciousness uh so that the person
was asking uh

if you know at least in the meditation
techniques with the when you put the
attention on for example your bodily
signals there you have a top down
modulation of this uh of precision
racing but the top 10 modulation is on
low level predictions

so you're not putting top down on what
you think about your body is you're
actually doing the opposite you're
trying to forget about your uh previous
expectations about the body because
you're trying to have this
non-judgmental awareness about what's

going on in your experience

and then uh with this

um let's say uh less uh biased uh way to
attend your body you start piercing and
perceiving uh

um

bodily uh Sensations in your body or
yourself or other kind of experiences so

I would say

this increasing Precision on
somatosensory signals happens at the low
level of the hierarchy

um that then attenuates the sensory
consequences of this

um of the first year

um

and the math awareness comes uh when you
learn

the when you're I think when you start
having increased body awareness so you
you'll make more experience I think then
you start developing meta awareness
about how your body is actually

um

about higher representation of the body
of your body but in the moment when this
happens uh I think it really happens
that the low levels of personal beliefs
and then later on the longer time scales
then it starts becoming uh more abstract
and ingrained in your personal uh at the
personal level of your beliefs

that's how I stay but of course uh if

Simon has something to do with it um

yeah it's really open

I know I didn't I didn't want to comment

that I just selfishly erased my hand to

ask another question Yeah okay or Daniel

I don't know if you have an opinion on
that I'm happy to hear it as well
well certainly
resonated with uh this comparative
approach to studying the the diversity
of
methods and protocols we already have
for mind and body and attention
broadening the scope and just kind of
pointing putting darts in the space of
of what practices do people engage their
mind and body in and then how can that
territory be explored with also a
variety of theoretical models and how
does that relationship connect to the
kinds of empirical work that we might do
in a brain scanner or um with other
kinds of measuring
um and I think one thing I
I I
want to explore more is like in a
hierarchical nested cognitive model
which we often speak of ourselves and
other uh types of sophisticated entities
as
where's our awareness where's awareness
Matt picture and um
and and also to Simone's question about
um when we're moving
but we're expecting what we're moving
then how does the exteroceptive flow
like the visual field you might be very
familiar with walking and so you might
be able to proprioceptively kind of zero
out on your walking
but there's this extra receptive
um flow of novelty in the world
so I think that that how

um and then connected back to the comparative and staring at a candle or a blank screen or darkness and all these different methods

um

it's it's there's a lot that that

these questions bring up

yeah

thank you Daniel so I I have something

to add to that because I think it's

super important so um

so for me it's not so you know my

background is like in body cognition and

activists and everything and for me it's

really not at all surprising the fact

that many of these techniques have been

developed by people that kind of like

lived in closed communities right so

it's like you have a space where people

can you know meditate walking I don't

know and then they have this ritual they

wake up at this time and they do this

thing so you put yourself in a yeah it's

a ritual right like all the

to reach also so this means it's like as

I said it's like high predictability

right

um and then that's that's something

super super uh important yeah and even

if even in the

walking meditation you need to

um follow a certain path

um uh yeah so that's that's super

interesting

well

um in in the last uh five or ten minutes

I would love to hear from each of you

like what are your directions heading

forward and and what's the um
what's exciting to you and where would
you like to take this
maybe um Simone and Julia first and then
we'll kind of make our way back to the
authorship

I mean I mean I'm not not myself not
working on the project right but like
what I what I was really interested in
sort of extending this is trying to
understand

um because the way
the way those deep meditative states are
reported sometimes and the way
um people with depersonalization
disorder report their experience is very
similar they almost did like the same
words right and to describe how they
feel and I wonder why one is experienced
positively and one is experience so
negatively as in it being a disorder and
completely disrupting someone's life
so that was like this this comparison I
think I would be very uh curious about
awesome Julia if you want to add
anything

um we're not it's it's very interesting
and um yeah I mean again I'm not so
involved in this study so it's a bit
hard but I mean I'm really interested in
the I mean all the Selfies affected by
yeah our interaction with the world
external world so uh and I'm completely
already spoiled with the result we have
with the international binding so um
uh it's very interesting I mean yeah
it's the self is attenuated or um if
it's disappear kind of I think that's

the key uh question we want to ensure
and uh yeah I would be very curious to
know I mean what very I found in uh data
uh to see if we can have a common
narrative about what's happening
um also with the personalization
um yeah it's very interesting to see how
moving or body I mean attention uh to
the body can really change the self kind
of for the way we process sales
information

um yeah so very interesting
or honor

yes so

um there's like lots of work to do ahead
but that's exciting right

um

yeah so very short term it's like we
need to write up this results that we
just got

um so yeah in relation to the body so
basically going to have some numbers to
show so that's that's super nice

um and then the next steps would be to
um you know

um

um

create or develop a computational model
around this yeah with basically contrast
you know the real world data with you
know um

simulation data and then uh yeah so have
a multi-layer type of like approach with
different methods from physiological to
behavioral to phenomenological to
computational potential also neural so
that's something with would be nice to
do in the next steps yeah

um so yeah

um and apart from that I think that's

all for now uh and I'll let uh I'll

leave uh the flow to Valeria

well again thank you again

and I know everybody uh yeah I I also

wanted to thank you a lot for this

platform this invitation it was yeah

really nice a really nice person and

really nice conversation

um

I'm working with on in these projects so

she already said what what's the the

next Direction on what I can add in this

I think would be

on the theoretical side also start

engaging with

um

more closely to bodily reactions to

meditation because the body boundaries

the solution is just one part of it it's

like one part of the phenomenology but

it's not the only thing that happens

during meditation

um there is a lot of potential that can

be discovered uh on this because

um

because the body boundary is the

solution again it's really study a lot

but it's a phenomenon and there is so

many other things that happens in our

body while we meditate and you I I speak

for first first person experience but in

the community it's really like already

sad other kinds of also um uh more

healing Traditions like breath work for

example

there there is a lot of potential where

people already work on that but there is
no Bridge uh with the academic
communities so people work pragmatically
with this intuitive knowledge as well
what happens in the body during these
alternate States Of Consciousness but
still we need to rage it with uh trying
to make sense from a phonological and
scientific point of view an experimental
point of view as well so I think I see
myself in yeah trying to step in this
tour then like bring Insight from one
world to the other and
opening uh to help myself and the other
people around me yeah
awesome well
it's the Vlog with the work that sounds
really exciting love the idea of the the
two-way street with different bodily and
cognitive practices and
collaboration and conversation that can
bring in all these different expertises
that that maybe don't have an fmri study
already
whatever that's worth
thank you though always um any of you
are welcome back and really appreciate
that you all came out
thank you Daniel
thank you see you soon